Pranjal Sapkota

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CS undergrad with hands-on experience building agentic AI systems, training LLMs from scratch, and deploying ML models to production. Interested in AI/ML engineering roles where I can work on real-world problems at the intersection of language models and applied machine learning.

Education

Vellore Institute of Technology, Vellore Expected May 2027
B.Tech in Computer Science & Engineering (Core) – CGPA: 8.48 / 10

Technical Skills

Languages: Python, SQL, JavaScript
Data Tools: Pandas, NumPy, Matplotlib, Seaborn
Agentic AI Frameworks: LangChain, LangGraph, LangSmith, MCP
ML/DL: PyTorch, scikit-learn, XGBoost, LightGBM
Web Development: Django, Streamlit, Tailwind, Next.js, FastAPI
Deployment: Railway, Render, Vercel, AWS S3
Developer Tools: Git, Docker, Linux, Google Colab, Kaggle, Google OAuth
Databases: MongoDB, PostgreSQL, SQLite

Research Experience

FaultBench-Industrial: A Standardized Fault-Injection Suite for Evaluating Self-Healing Strategies in Multi-Agent Industrial Control
Independent Research | Submitted to IEEE Access | Github | Website *Aug 2026*

- Built a benchmark that injects 5 fault types into a two-agent PPO control system and compares 4 recovery strategies, including a learned supervisor and an LLM replanner.
- Evaluated across 8 seeds with confidence intervals and effect sizes; found no single strategy dominates, and leaving severe faults uncorrected caused ∼17× more safety violations than rule-based recovery.

Comparative Evaluation of CNN and Vision Transformer Architectures for Waterborne Pathogen Detection from Plate Images
Faculty Guided Research | Submitted to IEEE Access | Review pending *Jan 2026*

- Evaluated plate images of 4 waterborne pathogens E. coli, K. pneumoniae, P. aeruginosa and S. aureus with CNN models (ResNet and EfficientNet) and Vision Transformer models (Swin T and ConvNeXt).
- Finding their model performances and evaluating robustness and generalization capabilities across different datasets and conditions.

Projects

FluxApply | Github | Website July 2026

- AI-powered web app that tailors resumes and cover letters to job descriptions in seconds, built with FastAPI, LangGraph, PostgreSQL and Next.js. Focus is solely on the anti-fabrication mechanism implemented.
- Designed an 8-node LLM pipeline with ID-based fact matching to rewrite only relevant resume content, integrated Google OAuth, AWS S3 storage, and Razorpay billing for a production-ready SaaS product.

MCP Agent | Github | Demo Jun 2026

- Built a ReAct agent loop in LangGraph with conditional tool-calling edges across web search, filesystem, GitHub, and Gmail via MCP.
- Persisted multi-turn conversation history via async SQLite checkpointer; Streamlit frontend with live streaming and tool-status indicators.

SmallLLM – Pathology Language Model | Github | Dataset Jun 2026

- Built a GPT-2 style decoder-only transformer in PyTorch (10.8M and 25M params) with self-attention, residual connections, and layer norm from scratch.
- Curated 52M characters from 13 pathology textbooks, cleaned OCR artifacts, published dataset on HuggingFace (with 59 public downloads); achieved val loss 0.8794 (25M model).

Credit Card Fraud Detection | Github | API | Frontend Mar 2026

- Built a stacked ensemble (RF, XGBoost, LightGBM, LR) on highly imbalanced data using class weighting and undersampling.
- Achieved F1 of 0.85 (Precision: 0.92, Recall: 0.79) after threshold tuning; deployed as a FastAPI endpoint on Render.

Certifications & Achievements

- Intermediate ML, ML Explainability, Data Cleaning, Pandas, Intro to ML – Kaggle, 2026
- Stanford University's Code in Place, 2025
- Best Freshers Award – InnovationX Hackathon organized by YANTRA, 2024
- Responsive Web Design – Free Code Camp, 2023